

SRVM — Signal Relevance Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Basis Standard (PBAS)

Operational Layer: Prediction Basis Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Signal Relevance

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Signal Relevance

Canonical Definition

Signal Relevance Module (SRVM) defines the governance conditions under which identified Prediction Inputs and predictive signals are assessed for their material relevance to the governed Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, Predictive Boundary, applicable Prediction Methodology, and relevant predictive conditions.

SRVM establishes whether an informational element has a legitimate predictive relationship to the prediction being formed.

It distinguishes:



information that is present

from:

information that is potentially informative

and:

information that is sufficiently relevant to participate in Prediction Basis.

SRVM does not determine whether the overall Prediction Basis is sufficient, whether the underlying information is evidentially valid, or whether a relevant signal ultimately improves predictive accuracy.

Its role is narrower:

a signal must first be relevant before it can legitimately contribute to Prediction Basis.

Operational Role

SRVM serves as the third internal governance space of the Prediction Basis Standard (PBAS) .

It receives:

Identified Prediction Inputs

and:

Governed Predictive Source Map

from the preceding PBAS modules.

The module asks:

Which identified inputs and signals are materially relevant to this specific prediction, why are they relevant, and under what predictive conditions does that relevance hold?

The governed progression is:

Identified Prediction Inputs + Governed Predictive Source Map → Relevance Criteria → Signal-to-Prediction Relationship → Relevance Determination → Governed Relevant Signal Set

The resulting relevant signal set becomes an input to:

ISUM — Input Sufficiency Module.

Internal Module Flow

Prediction Input Identification → Predictive Source Mapping → → Signal Relevance → Input Sufficiency → Prediction Basis Traceability

Current Position: Signal Relevance

Module Operational Space

SRVM governs:

signal identity; signal relevance; relevance criteria; Prediction Object relevance; Predictive Purpose relevance; Prediction Target relevance; Prediction Horizon relevance; Predictive Boundary relevance; methodology relevance; contextual relevance; temporal relevance; direct relevance; indirect relevance; conditional relevance; derived relevance; signal-to-outcome relationship; signal-to-condition relationship; signal-to-method relationship; relevance strength; relevance confidence; relevance uncertainty; relevance stability; relevance decay; historical relevance; current relevance; contextual shift;

regime change; redundant relevance; apparent relevance; spurious relevance; outcome-proximate signals; information-advantaged signals; excluded irrelevant inputs; relevance reassessment; and relevance traceability.

Module Function

SRVM prevents organizations from treating all available or identified information as equally meaningful merely because it enters a predictive environment.

A predictive system may have access to thousands of variables.

Many may be:

weakly related; historically correlated but no longer relevant; redundant; context-specific; temporally stale; indirectly related; dependent on the same underlying phenomenon; or only apparently predictive.

SRVM creates the governance layer between:

“the system used this information”

and:

“this information is materially relevant to the prediction.”

Input Presence $\neq$ Predictive Relevance

Predictive Signal

A Predictive Signal is an informational element whose relationship to the predicted outcome, predictive conditions, or methodological operation may materially inform the prediction.

A signal may be:

direct; indirect; quantitative; qualitative; observed; derived; model-generated; human-provided; simulation-derived; market-derived; contextual.

A signal is not automatically relevant merely because it is detectable.

Signal Identification and Relevance

PIIM identifies whether information participates as Prediction Input.

SRVM determines whether that input has a legitimate predictive relationship to the governed prediction.

This distinction is foundational:

Prediction Input $\neq$ Relevant Prediction Signal

An input may be present and still be irrelevant.

Relevance Criteria

SRVM requires relevance criteria to be sufficiently identifiable.

Criteria may relate to:

the Prediction Object; the Predictive Purpose; the Prediction Target; the Prediction Horizon; the Predictive Boundary; applicable methodology requirements; known conditions; relevant dependencies; context; timing.

A signal should not be considered relevant solely because it historically appeared alongside the predicted outcome.

Prediction Object Relevance

A signal must possess a meaningful relationship to the Prediction Object.

Information relevant to one object may be irrelevant to another, even where both exist in the same operating environment.

Environmental Presence $\neq$ Object Relevance

Predictive Purpose Relevance

A signal may be relevant for one Predictive Purpose and irrelevant for another.

For example, an indicator useful for broad strategic forecasting may not be sufficiently relevant for a near-term operational prediction.

Relevance must therefore remain purpose-specific.

Prediction Target Relevance

Signals should be assessed against the actual Prediction Target.

A signal capable of indicating a broad trend may not be relevant to the exact target being predicted.

Related Phenomenon $\neq$ Target-Relevant Signal

Prediction Horizon Relevance

Signal relevance may depend upon time horizon.

A signal useful for:

next hour

may be irrelevant for:

next year.

Likewise, structural long-term indicators may contribute little to immediate operational forecasting.

Relevant at Horizon A $\neq$ Relevant at Horizon B

Predictive Boundary Relevance

A signal may be relevant inside one geographic, organizational, demographic, operational, technical, or market boundary and irrelevant outside it.

SRVM therefore anchors relevance to the Predictive Boundary established under POPS.

Relevant Within One Boundary $\neq$ Universally Relevant

Methodology Relevance

A signal may only be relevant where the applicable Prediction Methodology can legitimately interpret or use it.

Information that has a conceptual relationship to the predicted outcome may still lack methodological relevance if the method has no valid mechanism for incorporating it.

Conceptual Relevance $\neq$ Methodological Relevance

Direct Relevance

A signal possesses Direct Relevance where it has a material and sufficiently identifiable relationship to the Prediction Target or the conditions immediately governing it.

Direct relevance does not automatically imply strong predictive power.

Indirect Relevance

A signal may influence the prediction through intermediate conditions or relationships.

For example:

Signal A $\rightarrow$ Condition B $\rightarrow$ Predicted Outcome

Indirect relevance should remain distinguishable from direct relevance where material.

Indirect Relevance $\neq$ Weak Relevance by Default

Conditional Relevance

Some signals are relevant only under specific conditions.

A signal may become informative only when:

another variable exceeds a threshold; a particular operational state exists; a market regime is active;

a system is operating in a defined mode; a specific event has already occurred.

Conditional relevance should not be represented as unconditional relevance.

Conditional Relevance $\neq$ Universal Relevance

Contextual Relevance

Contextual information may not directly predict the target but may determine how other signals should be interpreted.

Examples include:

regulatory state; weather regime; operating mode; market regime; geopolitical condition; system configuration; environmental state.

Contextual relevance can be material even where the information is not a direct predictor.

Temporal Relevance

A signal may lose relevance as time passes.

Historical relevance does not guarantee current relevance.

Historical Relevance $\neq$ Current Relevance

SRVM therefore requires temporal relationship to remain governable where material.

Relevance Window

Some signals have a bounded period during which they remain materially useful.

A Relevance Window may be defined by:

time; state; event; operational condition; regime; dependency.

A signal outside its relevance window may remain historically important while no longer contributing legitimately to the current Prediction Basis.

Signal Relevance Decay

Signal Relevance Decay occurs where a signal that was previously relevant becomes materially less relevant due to changing time, context, system state, or structural conditions.

Potential causes include:

stale information; changing market structure; behavioral change;

technology change; policy change; environmental change; altered operational conditions.

Valid Historical Signal $\neq$ Valid Continuing Signal

Regime Change

A signal may lose or change relevance when the system moves into a materially different regime.

Examples include:

normal market → crisis market; normal operations → emergency operations; stable system → degraded system; historical regulation → new regulation.

SRVM requires the possibility of changed relevance to become governable.

Signal Stability

Signal Stability concerns whether the relationship supporting relevance persists across the predictive conditions for which the signal is used.

A stable signal is not necessarily strong.

A strong historical signal is not necessarily stable.

Signal Strength ≠ Signal Stability

Relevance Strength

SRVM may represent the degree to which a signal is considered materially connected to the predictive context.

Relevance strength may be expressed qualitatively or quantitatively where methodology supports it.

Potential states may include:

Weak

Moderate

Strong

Condition-Dependent

However, relevance strength does not establish predictive accuracy.

Relevance Confidence

Organizations may possess differing levels of confidence in their relevance determination.

SRVM permits governance of uncertainty about whether a signal is actually relevant.

Relevance Determination ≠ Certainty of Relevance

This is particularly important in novel, sparse, or rapidly changing environments.

Relevance Uncertainty

Where signal relevance cannot be confidently established, the uncertainty should remain visible.

A signal need not be forced into a binary relevant/irrelevant classification where evidence does not support that precision.

Potential states may include:

Relevant

Conditionally Relevant

Potentially Relevant

Relevance Uncertain

Not Relevant

Apparent Relevance

A signal may appear relevant because it historically occurs alongside the predicted outcome.

Apparent relevance may result from:

shared causes; temporal coincidence; selection effects; duplicated underlying information; structural dependence; outcome proximity.

SRVM therefore prevents observed association from automatically becoming governed relevance.

Observed Association $\neq$ Governed Relevance

Spurious Relevance

A signal may appear predictive despite lacking a stable or meaningful relationship to the target.

SRVM does not require statistical doctrine or causal proof.

It requires that relevance claims remain sufficiently justified and bounded.

Historical Correlation $\neq$ Predictive Relevance by Default

Outcome-Proximate Signals

A signal may become highly informative because it appears very close to the moment at which the outcome becomes known or operationally determined.

Such a signal may still be relevant.

However, its relevance must not automatically be interpreted as demonstrating superior predictive capability.

Outcome-Proximate Information $\neq$ Superior Predictive Skill

PSMM preserves the underlying Predictor–Reality Proximity context.

Information-Advantaged Signals

Information available only to selected predictors may possess high relevance.

Its restricted availability does not make it irrelevant.

But predictive performance based on such information must remain distinguishable from methodological skill.

High-Relevance Private Information $\neq$ Superior Prediction Methodology

Redundant Relevant Signals

Two or more signals may each be relevant while expressing substantially the same underlying information.

SRVM may identify redundancy where it materially affects interpretation.

Redundancy does not necessarily remove relevance.

It prevents duplicated information from appearing as multiple independent confirmations.

Multiple Relevant Signals $\neq$ Multiple Independent Signals

Conflicting Relevant Signals

Prediction Basis may contain multiple relevant signals pointing toward different outcomes.

SRVM does not require one relevant signal to eliminate another merely because they conflict.

Where material, conflict should remain visible.

Relevant Signal A + Relevant Signal B $\neq$ Required Agreement

The methodology may govern how conflicting signals are processed.

Negative Relevance

A signal may be relevant because its absence, reduction, reversal, or contradiction materially informs the prediction.

Relevance is not limited to positive association.

Examples include:

expected event not occurring; signal disappearance; decreasing activity; missing confirmation; reversed trend.

Signal Exclusion

Signals determined not to be relevant should not remain silently embedded in the governed Prediction Basis merely because they are technically available.

SRVM may produce an exclusion or non-relevance state.

Exclusion does not imply that the information is false.

Irrelevant Information $\neq$ False Information

Relevance Reassessment

Signal relevance may require reassessment where:

Prediction Object changes; Purpose changes; Target changes; Horizon changes; Boundary changes; methodology changes; source conditions change; operational context changes; structural relationships change; new information emerges.

A previous relevance determination must not be treated as permanent by default.

Signal Relevance Traceability

A governed relevance determination should allow reconstruction of:

which signal was assessed; against which prediction; under what conditions; why it was considered relevant; whether relevance was conditional; what limitations applied; whether relevance later changed.

This prevents retrospective reinterpretation of signal importance after outcomes become known.

Retrospective Relevance Drift

After an outcome occurs, analysts may reinterpret previously ambiguous information as obviously important.

SRVM prevents later outcome knowledge from silently changing the governed relevance state that existed at prediction time.

Outcome Knowledge $\neq$ Historical Signal Relevance

Signal Relevance Sufficiency

Signal Relevance is sufficiently governed when:

material signals are identifiable; relevance criteria are established; relevance is anchored to the predictive frame; relevance conditions remain visible; direct and indirect relevance are distinguishable where material; temporal and contextual relevance are considered; relevance uncertainty is preserved; redundant and conflicting relevant signals remain identifiable; historical relevance is not retroactively rewritten; material changes can trigger reassessment.

SRVM determines signal relevance.

It does not determine whether the complete set of inputs is sufficient.

Governance Requirements

SRVM requires that:

1. identified Prediction Inputs be assessed for material relevance; 2. relevance be anchored to the governed predictive frame; 3. Prediction Object relevance be established where applicable; 4. Predictive Purpose relevance be considered; 5. Prediction Target relevance be considered; 6. Prediction Horizon relevance be considered; 7. Predictive Boundary relevance be considered; 8. methodology relevance remain distinguishable from mere conceptual relationship; 9. conditional relevance not be represented as unconditional; 10. temporal relevance remain governable where material; 11. historical relevance not automatically imply current relevance; 12. Signal Relevance Decay be identifiable; 13. regime changes trigger reassessment where material; 14. observed association not automatically establish governed relevance; 15. historical correlation not automatically establish predictive relevance; 16. information advantage not be mistaken for predictive capability; 17. redundant signals not be mistaken for independent confirmation; 18. conflicting relevant signals remain visible; 19. relevance uncertainty remain explicit where material; 20. historical relevance determinations remain traceable.

Governance Logic

SRVM applies the following governance logic:

Identified Prediction Inputs + Governed Predictive Source Map

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Relevance Criteria Established

↓

Prediction Object / Purpose / Target / Horizon / Boundary Alignment Assessed

↓

Methodology Relationship Assessed

↓

Direct / Indirect / Conditional / Contextual Relevance Identified

↓

Temporal Relevance Assessed

↓

Source and Predictor–Reality Proximity Context Considered where material

↓

Redundancy / Conflict / Apparent Relevance Assessed

↓

Relevance Uncertainty Determined

↓

Signal Relevance State Established

↓

Governed Relevant Signal Set

↓

ISUM — Input Sufficiency Module

Where relevance cannot be sufficiently established:

Signal Relevance Governance Gap → Clarification / Restriction / Exclusion / Reassessment

Minimum Implementation Framework

A minimum implementation of SRVM must establish:

A. Signal Reference

The Prediction Input being assessed.

B. Prediction Reference

The governed prediction or predictive frame to which relevance is assessed.

C. Relevance Basis

Why the signal is believed to possess a material relationship to the prediction.

D. Relevance Dimension

Where applicable:

Object / Purpose / Target / Horizon / Boundary / Methodology / Context / Condition

E. Relevance Type

Where applicable:

Direct / Indirect / Conditional / Contextual / Derived

F. Relevance State

At minimum:

Relevant / Conditionally Relevant / Potentially Relevant / Relevance Uncertain / Not Relevant

G. Temporal State

Where material:

Current / Historical / Decaying / Expired / Unknown

H. Relevance Strength

Where used:

Weak / Moderate / Strong / Condition-Dependent / Unable to Determine

I. Source Context

Relevant source dependency or Predictor–Reality Proximity context from PSMM.

J. Redundancy State

Whether the signal materially duplicates information already represented by another signal.

K. Conflict State

Whether the signal materially conflicts with other relevant information.

L. Reassessment Trigger

Conditions requiring relevance review.

M. Historical Preservation

The relevance state applicable at prediction formation must remain reconstructable.

Governance Outputs

SRVM may produce:

Governed Relevant Signal Set; Signal Relevance Register; relevance criteria; relevance-state determination; conditional-relevance record; temporal-relevance state; Signal Relevance Decay indicator; relevance-strength record; relevance-uncertainty state; redundant-signal record; conflicting-signal record; excluded irrelevant-input record; relevance reassessment trigger; historical relevance record; unresolved relevance requirements.

The primary governed output is:

Governed Relevant Signal Set

This becomes a principal input into:

ISUM — Input Sufficiency Module.

Critical Distinctions

Input Presence $\neq$ Predictive Relevance

Prediction Input $\neq$ Relevant Prediction Signal

Environmental Presence $\neq$ Object Relevance

Related Phenomenon $\neq$ Target-Relevant Signal

Relevant at Horizon A $\neq$ Relevant at Horizon B

Relevant Within One Boundary $\neq$ Universally Relevant

Conceptual Relevance $\neq$ Methodological Relevance

Indirect Relevance $\neq$ Weak Relevance by Default

Conditional Relevance $\neq$ Universal Relevance

Historical Relevance $\neq$ Current Relevance

Valid Historical Signal $\neq$ Valid Continuing Signal

Signal Strength $\neq$ Signal Stability

Relevance Determination $\neq$ Certainty of Relevance

Observed Association $\neq$ Governed Relevance

Historical Correlation $\neq$ Predictive Relevance by Default

Outcome-Proximate Information $\neq$ Superior Predictive Skill

High-Relevance Private Information $\neq$ Superior Prediction Methodology

Multiple Relevant Signals $\neq$ Multiple Independent Signals

Relevant Signal A + Relevant Signal B $\neq$ Required Agreement

Irrelevant Information $\neq$ False Information

Outcome Knowledge $\neq$ Historical Signal Relevance

Enterprise Application

SRVM enables organizations to reduce a common predictive-governance failure: confusing information volume with predictive meaning.

Modern prediction systems can ingest enormous quantities of data, external content, AI-generated analysis, operational signals, market information, and human judgment.

Without Signal Relevance governance, organizations may:

use stale variables; preserve irrelevant features; overvalue historically correlated signals; treat redundant information as confirmation; mistake outcome proximity for predictive skill; retain signals whose relevance disappeared after a regime change.

SRVM establishes a controlled relevance layer before PBAS asks whether the overall informational foundation is sufficient.

Use Case 1 — A Historically Powerful Signal Stops Working After the Environment Changes

A retailer has used regional mobility data for several years to predict short-term store demand.

Historically, mobility and demand were strongly aligned.

A major shift toward home delivery, remote work, and changed shopping behavior alters the relationship.

The mobility signal remains available and continues to enter the model because it has historically been one of the strongest predictive variables.

Its historical performance causes the organization to continue treating it as highly relevant without reassessment.

The critical governance question is:

Can a historically powerful signal remain a governed relevant signal when the operating conditions that created its predictive relationship have materially changed?

SRVM requires relevance to be reassessed when contextual or structural conditions change.

The signal may remain relevant, become conditionally relevant, weaken, or cease to be relevant.

Historical Relevance $\neq$ Current Relevance

Use Case 2 — A Predictor Looks Brilliant Because the Signal Appears Immediately Before the Outcome

A forecasting system predicts whether industrial equipment will fail.

One input consistently provides extremely strong predictive performance: an internal maintenance alarm that appears shortly before failure.

The organization begins presenting the system as possessing exceptional long-range predictive capability.

However, the alarm is generated only after the equipment has already entered a failure state that operational technicians recognize as highly likely to result in shutdown.

The critical governance question is:

Does the strong relationship between the alarm and the outcome demonstrate superior predictive insight, or is the signal highly relevant because it occurs extremely close to the emerging Operational Reality of the failure?

SRVM recognizes the signal as potentially highly relevant while preserving its temporal and operational context.

PSMM preserves the source's proximity to Operational Reality.

Later PGA™ performance governance can then avoid confusing:

high signal relevance

with:

deep predictive capability.

Outcome-Proximate Information $\neq$ Superior Predictive Skill

Canonical Boundary

SRVM governs the determination, qualification, conditionality, temporal stability, uncertainty, reassessment, and traceability of Signal Relevance within a governed Prediction Basis. It does not govern Prediction Input identification, detailed source mapping, Prediction Basis sufficiency, evidence validity, predictive assumptions, prediction formation, probability representation, prediction feasibility, Operational Reality, collective prediction legitimacy, or predictive performance.

SRVM establishes:

which identified inputs legitimately matter to the prediction.

It does not determine:

whether the resulting Prediction Basis is sufficient to support the prediction.

Canonical Closing Statement

Information becomes part of a governed Prediction Basis not because it is available, measurable, or historically associated with an outcome, but because its relevance to the specific prediction can be sufficiently established.

SRVM separates informational presence from predictive meaning before PGA™ determines whether the Prediction Basis is sufficient.

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